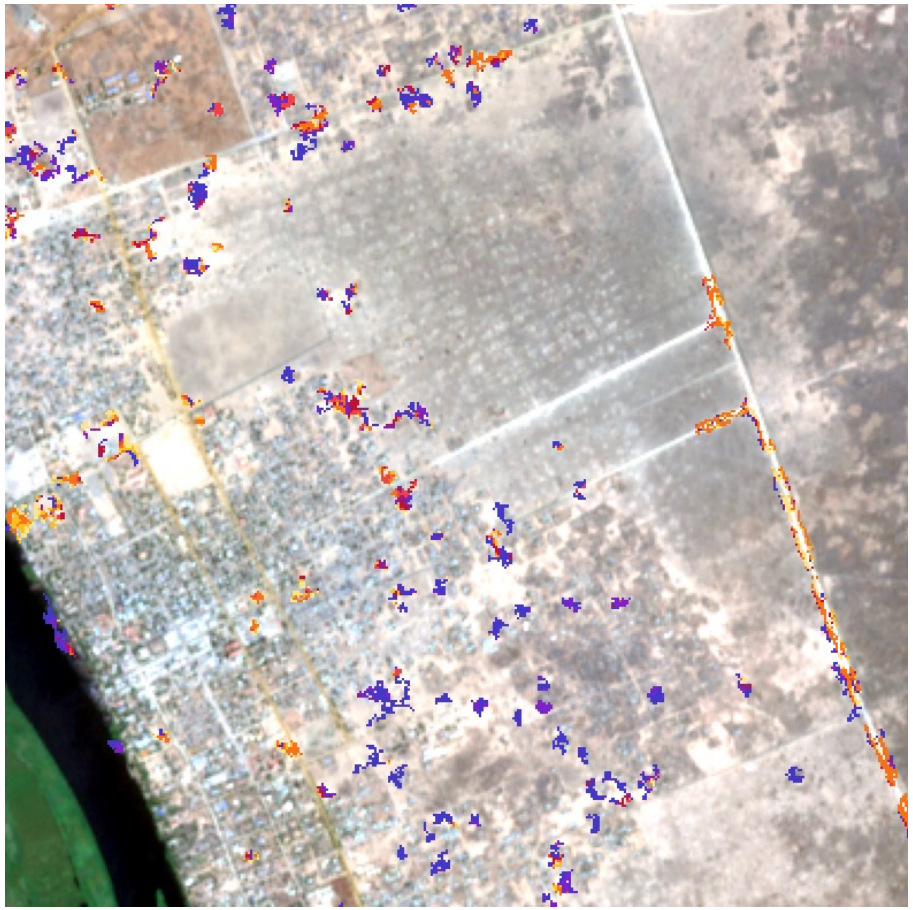
Built-up change, on demand

From static global layers to city requests

A proof of concept for on-demand, agent-ready products

October 2026

Cristiano Giovando



Malakal, South Sudan: new built-up by dry season, 2020-2026

Three questions a static layer cannot answer

Where is it growing?

Rank the fastest-growing cities and neighbourhoods

Chatbots and one-button maps

Since when?

Date each new built-up area to the season it appeared

Building inventories need age

How sure are we?

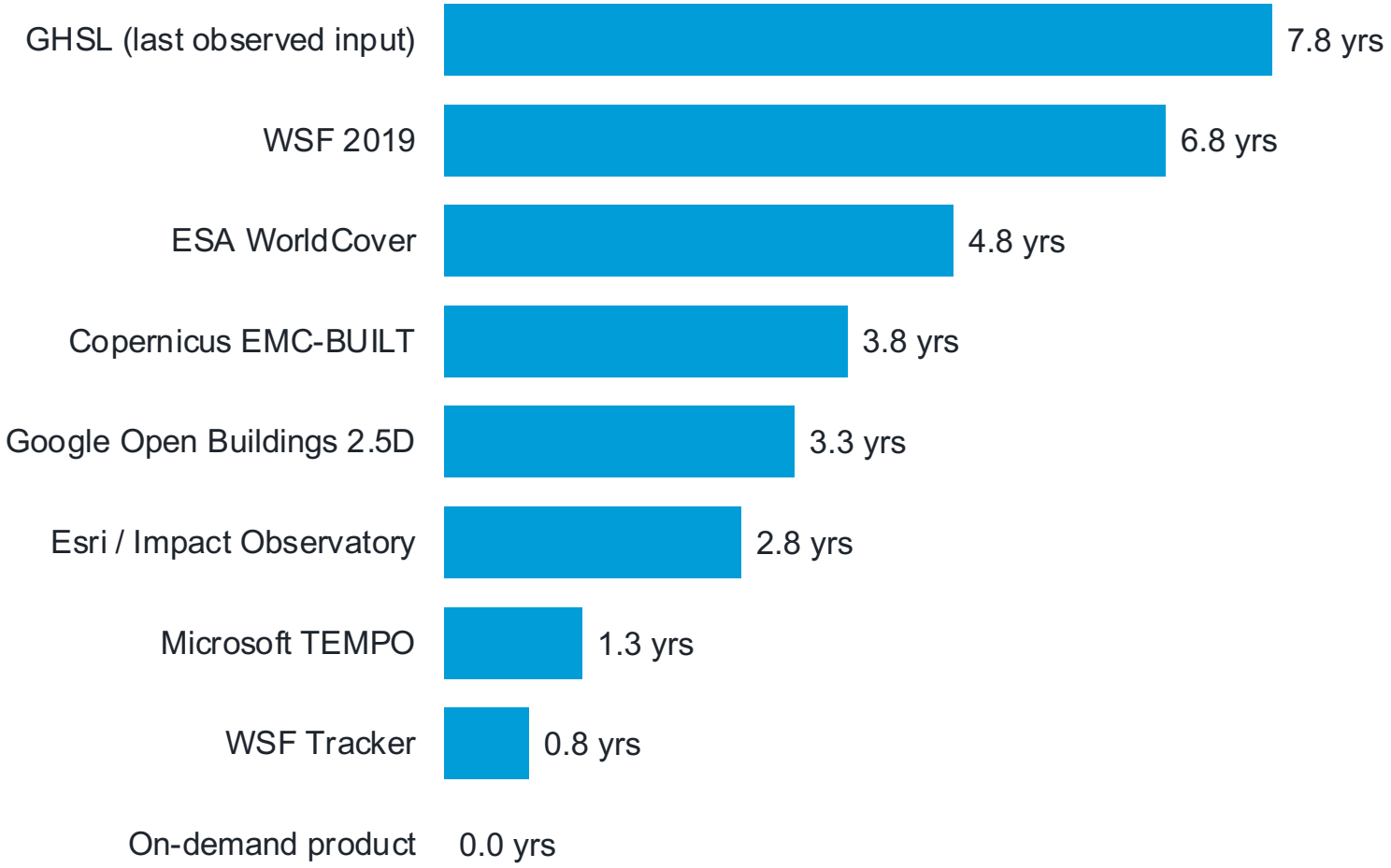
State accuracy for this city, not a global average

Every figure behind a decision

Fast-growing and crisis cities (Malakal, Renk, Juba) change faster than any annual layer is refreshed.

Static layers describe cities as they were years ago

Age of the newest observation in each layer on 28 October 2026



Newer is not observed

GHSL 2020-2030 epochs are modelled from a 2018 observation.

The freshest open layer

WSF Tracker: data to Jan 2026, published Jun 2026.

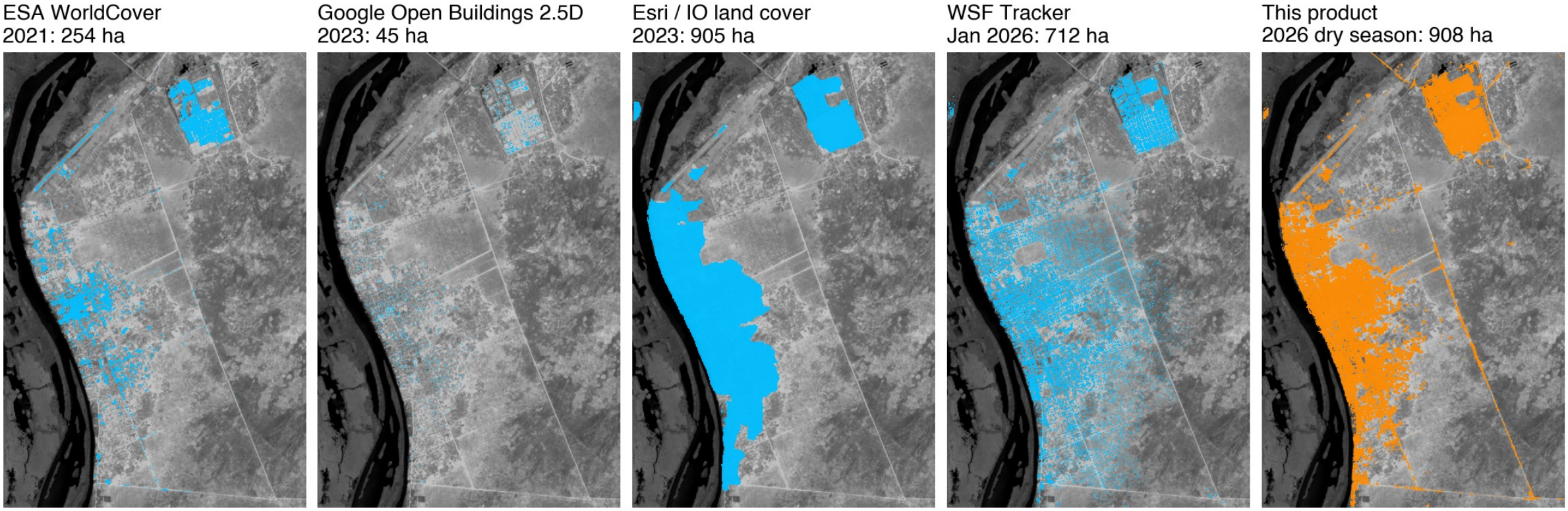
On demand

Any city, on last week's imagery.

End of each product's newest observed period. Sources: JRC GHSL R2023A, DLR WSF, ESA WorldCover, EMC-BUILT R2025A, Google Open Buildings Temporal, Planetary Computer io-lulc-annual-v02, Microsoft TEMPO (arXiv 2511.12104), WSF Tracker (source.coop).

Same town, same grid: the layers disagree twentyfold

Built-up area in one 66 sq km area over Malakal, South Sudan, newest epoch of each layer

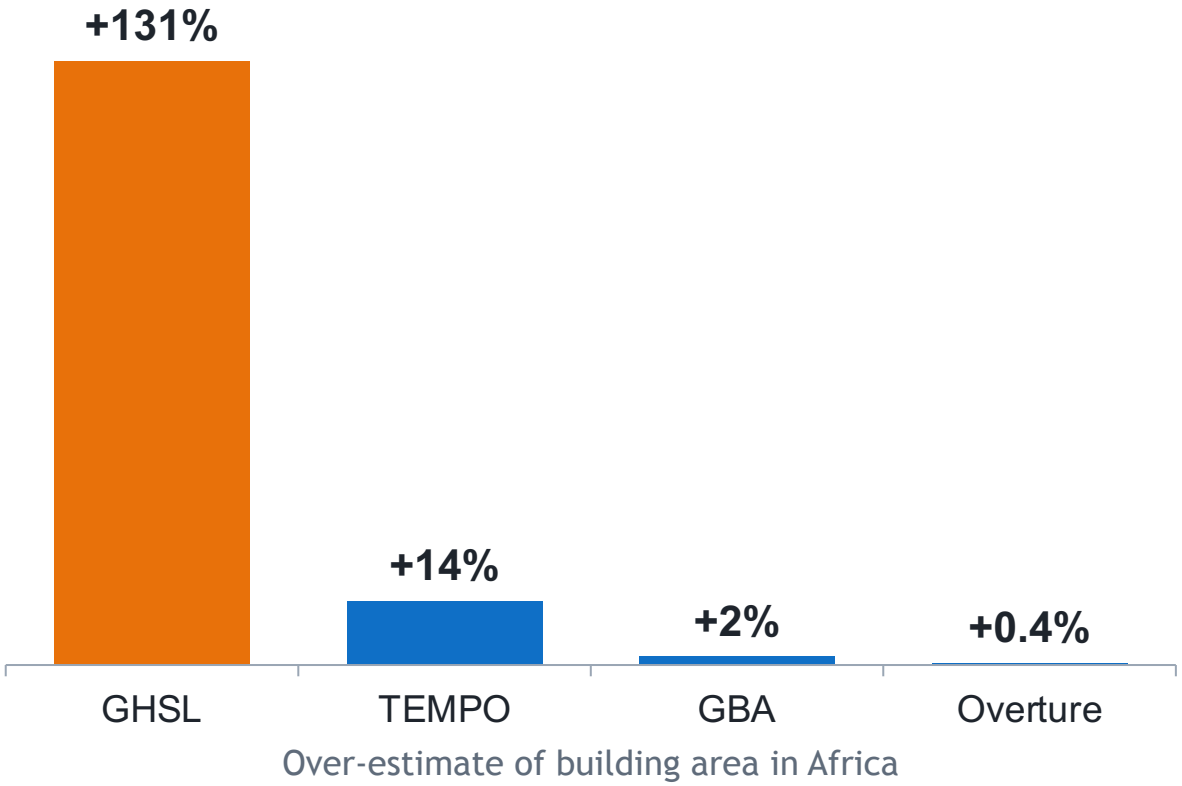


45 ha to 905 ha, with no city-level accuracy statement. Any answer depends on which default was picked.

All layers read on the same 10 m grid from cloud-native endpoints. 'This product' is the Dec 2025 - Mar 2026 dry-season composite.

Accuracy is rarely published, and worst in Africa

Independent benchmark of global building layers against hand-labelled data, Africa



Over-estimate of building area in Africa

GHSL: R^2 of -0.37 in Africa

Worse than assuming the same density everywhere.

Per-region accuracy is the exception

Microsoft footprints publish it; WSF Tracker and GHSL do not.

Licences matter too

GlobalBuildingAtlas heights are licensed for non-commercial use only.

An on-demand product can ship with its own accuracy statement for the requested city.

Lahrichi, Allabadi, Bradbury, Malof (2026), arXiv 2607.19766 (77 m grid). Microsoft GlobalMLBuildingFootprints; WSF Tracker README; GlobalBuildingAtlas README.

A city request becomes a product in minutes

builtup-mcp: open source, open data, cloud-native

1

Request

City, area, years and the season to compare

2

Composite

Cloud-free Sentinel-2 and Sentinel-1 radar, per dry season

3

Learn

Gradient-boosted trees trained on the city's own open footprints

4

Date

Each new built-up area tagged with the season it appeared

5

Deliver

A STAC item: maps, polygons, stats, accuracy

10 min

first run for Malakal (66 sq km)

1 min

re-run from cache

189

satellite scenes, all open data

0

licences or credentials needed

Code: github.com/cgiovando/builtup-mcp (Apache-2.0). Model: scikit-learn HistGradientBoostingClassifier, fused, optical-only and radar-only. Also considered: fixed index thresholds (v0.1), global classifiers (Dynamic World, WSF), deep segmentation and EO foundation models. Detail in METHODOLOGY.md.

Malakal: where and when new built-up appeared

Dry seasons 2019-2026, 10 m, from open data only

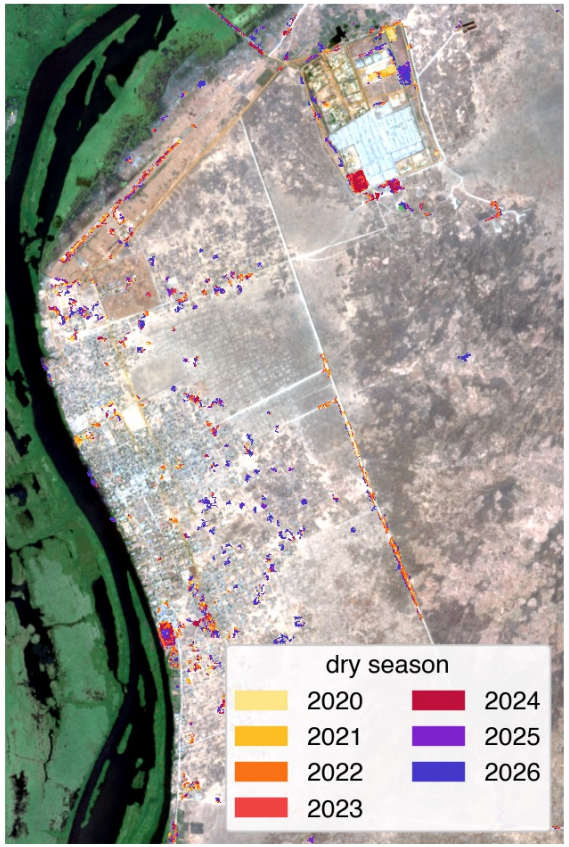
Sentinel-2, 2019 season



Sentinel-2, 2026 season



New built-up, first seen in



At the full request AOI (106 sq km): built-up 849 to 1,092 ha, 243 ha new, WSF extent agreement 0.73.

+19%

built-up, 760 to 908 ha

148 ha

new built-up, confirmed in dry seasons

105 ha

provisional, seen only in Jul-Oct 2026

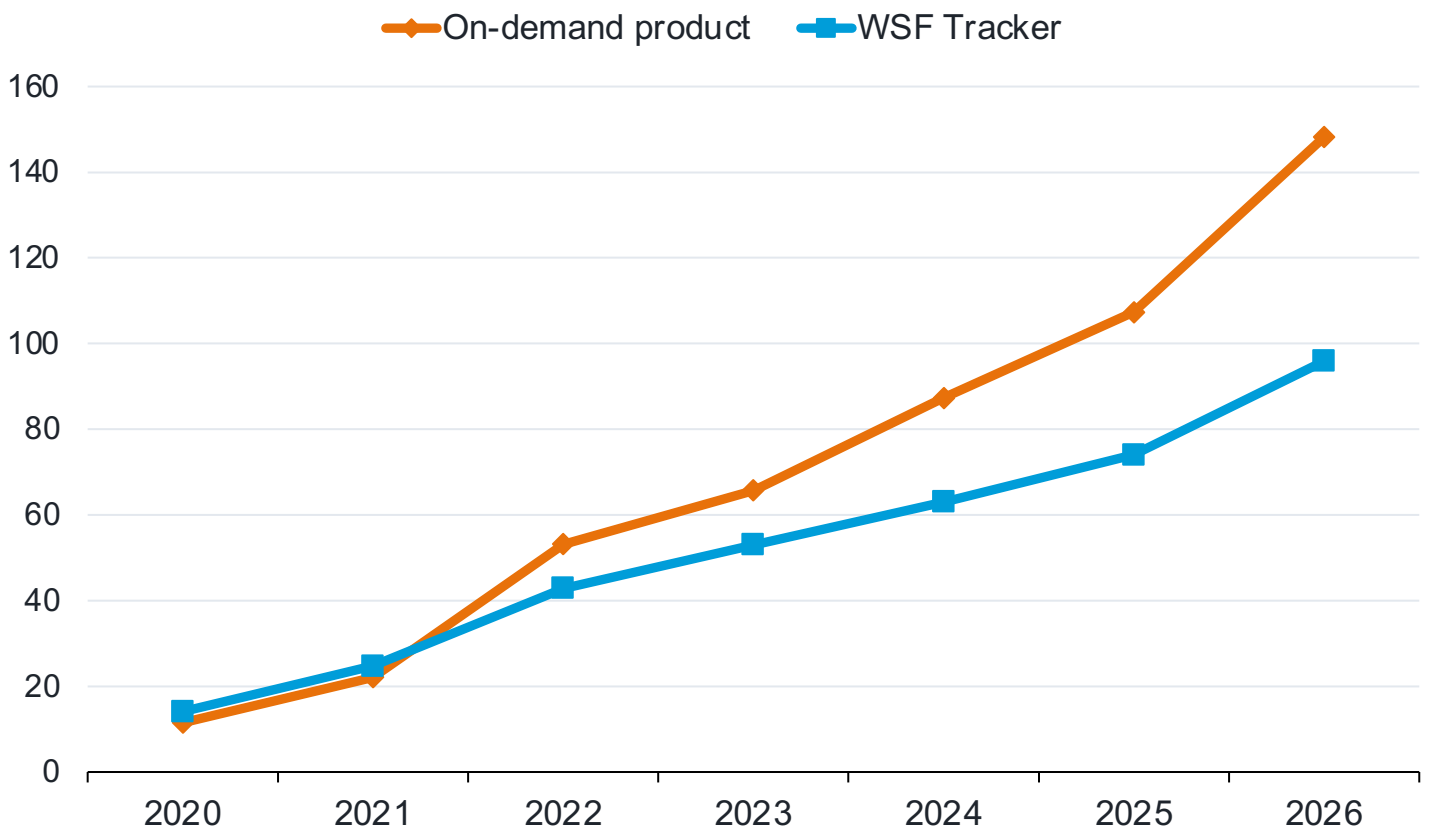
2022

median date of the open footprints

Sentinel-1 had fewer than 4 acquisitions per season over Malakal in 2022-2025; those seasons were scored on optical data only.

Checked against WSF Tracker: same town, same trend

Cumulative new built-up in the Malakal area since the 2019 dry season (ha)



0.756

agreement on where the town is (pixel F1, Jan 2026)

148 vs 96 ha

new built-up since 2019: this product vs WSF

0.211

agreement on which pixels are new

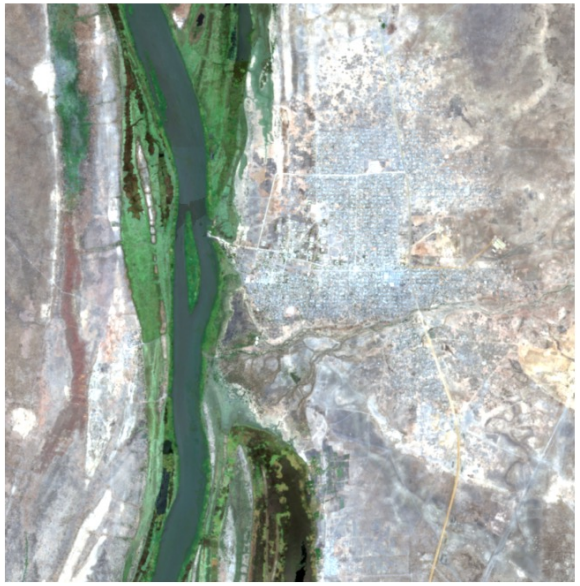
Independent methods agree on the town and the trend, not on every new pixel. A reference sample settles it, per request.

WSF Tracker GeoZarr (DLR / MindEarth), six-monthly epochs mapped to the following dry season; same 10 m grid, one-pixel tolerance.

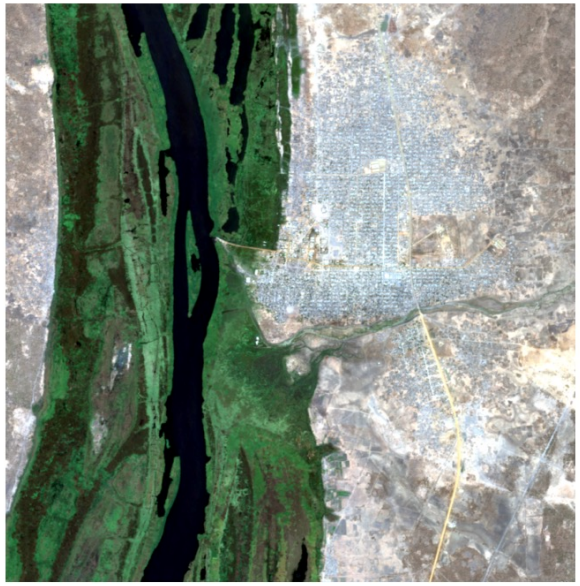
Renk: the border town where returnees arrived

Dry seasons 2019-2026, the same request template as Malakal

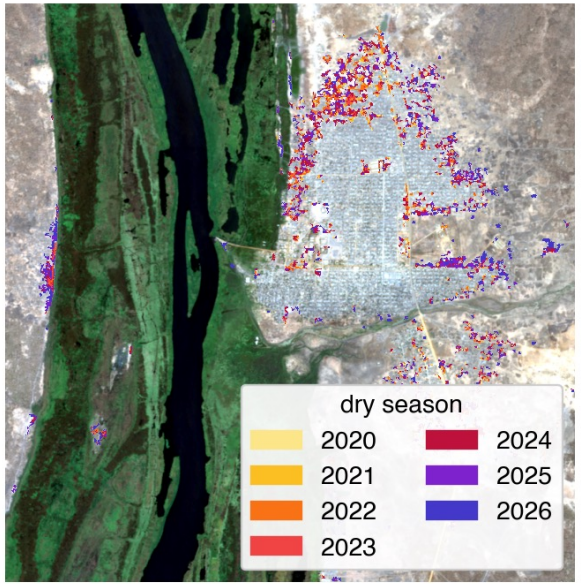
Sentinel-2, 2019 season



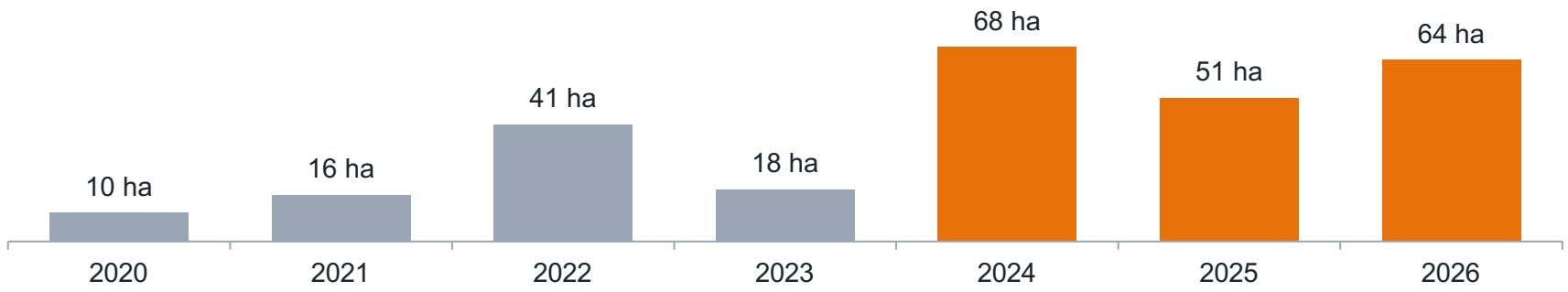
Sentinel-2, 2026 season



New built-up, first seen in



New built-up first seen in each dry season



+32%

built-up, 847 to 1,115 ha

268 ha

new built-up, confirmed in dry seasons

183 ha

of it first seen in 2024-2026 seasons

0.881

extent agreement with WSF Tracker

Context: REACH, Renk emergency assessment brief (Jul 2025): more than 800,000 people crossed from Sudan at Renk since Apr 2023, with built-up growth on the town's outskirts.

The explorer reads the products like any STAC client

Season slider, static-layer overlays, neighbourhood statistics



Maps

Built-up probability and season of change (COG)

Polygons

New built-up with season, area, confidence (GeoParquet)

Neighbourhoods

H3 statistics for ranking and the chatbot

Provenance

Scenes, model, accuracy, benchmarks (JSON)

Same item opens in QGIS, Python, R, Databricks or any STAC catalog.

Local demo: uv run builtup-demo. Products are STAC 1.1 items written with pystac; rasters are Cloud-Optimized GeoTIFFs.

An agent answers questions from the products

Real exchanges with the builtup-mcp server over the Model Context Protocol

"Which cities grew fastest?" compare_products

```
- Renk: built-up 846.7 -> 1114.9 ha (2019-2026), 4.0%/yr, new 268.2 ha
- Malakal (request AOI): built-up 848.8 -> 1092.1 ha (2019-2026), 3.7%/yr, new 243.3 ha
- Malakal: built-up 760.2 -> 908.3 ha (2019-2026), 2.6%/yr, new 148.1 ha
- Juba: built-up 4103.4 -> 4723.5 ha (2019-2026), 2.0%/yr, new 620.1 ha
- Kigali: built-up 4288.7 -> 4549.1 ha (2019-2026), 0.8%/yr, new 260.4 ha
```

"Where in Malakal did it grow most?" rank_neighbourhoods

```
Top 3 cells by gain_ha_total in Malakal:
- 886b5b64e3fffff (9.5108, 31.6608): new built-up 14.8 ha, built 37.7 -> 52.5 ha (2019 season...
- 886b5b66b9fffff (9.5576, 31.6792): new built-up 7.8 ha, built 5.0 -> 12.8 ha (2019 season -...
```

Every answer cites a product id, its date and its accuracy, because the tools read published products, not the open web.

Tools: plan_city_request, run_city_request, list_products, get_product_summary, rank_neighbourhoods, compare_products, static_layers_report.

One request template, four cities

Dry-season composites, 2019 to 2026

Malakal

66 sq km, Dec-Mar

2.6% / yr

built-up growth

148	0.756
ha new	WSF agreement

Radar gap: 2022-2025

Renk

77 sq km, Dec-Mar

4.0% / yr

built-up growth

268	0.881
ha new	WSF agreement

Radar gap: 2022-2025

Juba

66 sq km, Dec-Mar

2.0% / yr

built-up growth

620	0.967
ha new	WSF agreement

Radar gap: none

Kigali

69 sq km, Jun-Aug

0.8% / yr

built-up growth

260	0.964
ha new	WSF agreement

Radar gap: none

Renk, Juba and Kigali areas are illustrative boxes, not city requests. WSF agreement: pixel F1 on built-up extent at the matching date.

Where this fits among the alternatives

Others win on resolution, archive depth and building detail

	Newest data	Per request	Accuracy published
WSF Tracker	Jan 2026	No	No
GHSL	2018 observed	No	No
Google Open Buildings 2.5D	2023	On request to Google	Not for African heights
Microsoft TEMPO	Q2 2025	No	Subsets
Planet change feeds	Weekly	Subscription	Global only
Copernicus EMS	Per event	Via authorised user	Per product
builtup-mcp	**Last week**	**Self-service**	**Agreement, every product**

Planet, Vantor and Ecopia resolve individual buildings; WSF Tracker and TEMPO offer a harmonised global archive; EMS grades damage in about two days.

Demo, and what comes next

Live demo

- Malakal explorer: play the seasons, overlay the static layers
- Agent: fastest-growing neighbourhoods, then plan the next city
- Open the STAC item in QGIS

Next

- Reference sample for Malakal from VHR imagery
- Date existing footprint inventories by season of change
- NISAR L-band once validated (planned Q4 2026)

Know the limits

- 10 m built-up surface, not individual buildings
- Bright bare ground can read as built-up
- Proof of concept: not yet validated against an independent reference

The next city request is a JSON file and ten minutes: github.com/cgiovando/builtup-mcp

Full limitations and method in METHODOLOGY.md.

Thank you

github.com/cgiovando/builtup-mcp

Open source and open data: every figure in this deck can be reproduced from the request files in the repository.

